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SUMMARY KEYWORDS

OpenShift, command line, production, pipelining, architecture lab, CDE chain, security, zero risk profile, hardened images, project honey bird, software factory, RPMs, dependency trees, automation, API.

SPEAKERS

Speaker 1, Speaker 12, Speaker 6, Speaker 5, Speaker 4, Speaker 7, Speaker 8, Speaker 2, Speaker 11, Speaker 10, Speaker 3, Speaker 9

Speaker 1 00:00

Some command line things. We also have an entire section on how to bring this into OpenShift and do a little more pipelining things and some production stuff that you might do with OpenShift cluster. My name is Matt, my senior. We also have with us today the man, the myth, the legend, the product manager for our pages. Ben Green Art,

Speaker 2 00:22

I'm the guy who gets yelled at you

Speaker 1 00:24

and the author of our section,

Speaker 3 00:28

come on. Come on. Our

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Speaker 1 00:31

architecture lab itself. Here to hear about. So this is something that really give you a little background. This is the sort of things our teams are facing, before Methos, before all these weird AI things the CDE chain. CDE reporting has been accelerating. The flood is just now bigger. The sheer volume of things, developer, security, team, platform teams need to keep on top of will not stop growing from here. This report, reported something on the order of a 400% increase in reports in the past two years. More reports last year than we're sorry it was last quarter than the year before. Something It's whatever

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Speaker 4 01:37

you think it

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Speaker 1 01:38

is. It's worse. On top of all of this inbound information, we also have a very different zero risk profile getting pushed down on top of us from organizations. People are no longer satisfied with a long list of exemptions. They want a long list of nothing. So getting pushed harder and harder, harder to completely clear that national which means you're spending all of these cycles that should be spent on features and your customers and your application on trying to figure out who to blame for a package you didn't install on a container image you don't know. That's what we would like to try and solve. Enter redheaded images. This is a set of distro list. Shelves hold no shells, just that. The only thing they need is before that, whatever purpose it is, a runtime, a language, a server as small as we can possibly make them, built correctly in a hardened environment by Linux experts, available across the board wherever you want to use them at no cost, and delivered by folks who know how to deliver secure Linux rpms. These are the guiding principles we had during the project. Right minimization only. The Essentials is do the thing it needs to do. Make sure that we're doing all the right things. We're hardening. They come out of the gate in a state that is a step ahead for you. But none of this works if the developers won't use it. The more groups you put in front of people, make them secure, the less secure you'll actually end

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Speaker 4 03:29

up write that down.

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Speaker 5 03:31

Good security is simple security.



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Speaker 1 03:34

So in order to do these things, we need to make changes around the way we deliver software. This is not real. We cannot operate at this pace and continue to live in the same way that we deliver rel which means we're doing this by tracking upstream much, much, much more closely in order to get that extremely close window between publication of fix and delivery of new image. And we also want to make sure that you can do all of this and verify that it was done correctly, and that when you have this thing we said we built, when we say hard main, we're talking about everywhere from the very pipeline we use, which is a relatively new salsa, three highly automated thing that we automated even more we're doing all the things we've learned over 25 years of building software and packages with green compiler options by taking all the latest and greatest work that came Before us on what a distress image should be and which version of that we agreed with as the best approach to give the best experience, and then adding to that, the ability to create starting points for profiles that you all need in various different regulatory environments. So that way, we're starting with the most secure baseline that you can use that gets you the farthest ahead. Anybody can read the same bolt at the 100 meter dash, if I start you at 99 meters, and that's what I want to do. So the core of this is something we call we call project honey bird. We announced in November of last year. This is really sort of the Software Factory, right? This is, how do we build these? And we have this idea for images. How do we get there? If you want images, you need RPMs, because we're Red Hat, we're not going to completely change the way we deliver software. That's insane, but we will completely change how we deliver our PMS. Think this as a the most rolling release you have ever seen. This is not designed for the jump to be a general purpose distribution. It was designed to deliver a catalog of images so the dependency trees look different than you might see if you use in Fedora install the same rpm, same rpm in air quotes, because we're doing different things, so that when DNF installs things, it's not saying, oh, right, emacs needs all of Qt and Xlib. So that means we get these nice little statistics, and thereby those to see the images wind up 80 from 85% smaller. We wind up getting images down by 60% we're updating our images on a 20 424 hour target. That brings us to this mythical, near zero target. When I say mythical, I mean, actually verifiable on the website, and then it's going to walk you through it. This is the actual workflow for prod sec, our product security team. I'm not going to walk you through it. I just want you to know that is what the security process looks like. It's complicated. In order to do this, we need to be able to do it as quickly as any humanly possible, because people can't move this. We had to automate this our level in software engineering, as well as with product security and partnership to make sure we deliver the images and the security CPE feeds that you need with this 24 hour goal that we have in mind, goal, not SLA, so that we can actually hit all the benchmarks that we want to be able to do, deliver these images in ways that make them usable for you. We also don't want to force you into another model I mentioned. These are free to use. We don't care where you run them. We would love you to run them in OpenShift and on Red Hat Enterprise Linux, because those are the things we build, and we like them. If you want to run them in Eks, please do if you would like to support and they get a show of hands. How many people here have a rel subscription somewhere cool? How many people have an open chip subscription somewhere? Cool, everybody who raised a hand now call us on a hardened image and say, hey, my Python image isn't Python if it's your code, but if it's our container. Now we might ask, could you run it on OCP instead of Eks, just to see if it is a weirdness in something we did with the container? But we're not going to save them. It also means you can replace components out of your stack individually, and not have to try to re architect your entire application just to start taking advantage of these new images. So here's what you can go through, managing the walking through the new way we present all of this branding information. Go through usability, available some security workflows. How do you do? Build automation, secure pipelines, loaded two modules after you hear them talk. They are similar, but they do not build on each other. If you feel like you only want to see the OCP things,

jump to module two, just do that. Have a good day. Catch your light if you want to just do the CLI, pop, man, REL, stuff. Module One, do them in any order. Concepts build on each other, but there's no hard requirement to do them in any either particular order. Just do the steps in the module in the right order. Just pick one or the other. This is how you will get to it. This should be the homepage on your browser on all of your laptops you look for 2865 somewhere probably near the bottom,

 Speaker 6 09:46
because

 Speaker 1 09:46
that's a high number. Once you log in, you'll see our Red Hat images. Click on that access to lab, you'll be asked for your email, the email that used to

 Speaker 5 09:57
register

 Speaker 1 09:57
some of these little work here and today's password. Today's password is 00, 10. Once you get that, you'll see a interface that has some instructions on one side a bunch of tabs. Go ahead and navigate across the tabs. There'll be a thing that says around 100 pages that will bring you to our new page that is also just a web page. If you feel more comfortable just opening a new tab in your browser, you can do that. You go to red.hd/i word images, do Red Hat com, as you get ready, I'm going to pass the mic to them. You

 Speaker 4 10:43
something, hey, so these images are built. We have human interfaces, and then machine interfaces. So I'm going to show you the human one, right? Everybody here is human, right? And by the way, who can tell me where the name project hummingbird comes from? This guy knows I don't just get it. When we were having discussions around the name, there was one we couldn't figure it out for more time. I liked him, and there's one voice of wisdom in the room. They said, why don't we call it project? This guy right here. Thank you, man, yeah, you can.

 Speaker 1 11:28
You don't

 Speaker 4 11:28
have to definitely get and we do have stickers. If you want stickers, the T shirts are awesome, by the way. Thank you. Thank you. Thank you. Okay, so if you go to images that calm, and this is a race, so go to images.com zero. Hero 10, zero. Hero, one, zero. It's a little binary joke for you guys.

 Speaker 7 12:07
There are 10 types of people. What are we supposed to use for the email?

 Speaker 4 12:11
For the email should

 Speaker 1 12:12
be the one that you registered with, so whatever your email that you gave us, when you said

 Speaker 4 12:17
no spam at hotmail,

 Speaker 1 12:20
that's Yeah,

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Speaker 4 12:21

we all the same one. Okay, so I want you to find these things. Where's my full command? Count CBDs, count the variants. Find the s bomb and variant, right. Ready? Is everybody on the website. Nobody's on website. That's okay. Go to the website. One guy's nodding. One guy's on site. Yes. Do you guys have internet at home? Your computer? Come with dinner? Yeah. Okay. Got

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Speaker 8 12:55

an AOL CD.

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Speaker 4 12:56

AOL CD. You got some minutes? Yeah. Oh my god, the world has changed.

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Speaker 1 13:01

Okay,

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Speaker 4 13:02

look it up for Maria DB with the latest tag CDs. Oh yeah, there's the bottom part. What's the number of CVS was first? I saw you? Second, yeah, third, fourth, yeah, how many variants? Two, four, here's the next slide, all right. Show

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Speaker 1 13:43

them how to find it.

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Speaker 4 13:44

So on the beautiful website, you can see, all right, first thing I look at is my streams. These are the major versions that we packaged up. And then the very end, which is the image type. Default is like your small, minimal, digital list, one the builder, one has more utilities. It's all the same hardening, but you have a package manager if you need DNF install or to build to be able to debug. Use that. Here's your pull command and the current CVE count. This number will change over time. That's a good thing, right? This is accurate. All right, ready for another one? Real quick. Same thing, core runtime, FIPS and match the tag. Yes. One, CV, zero. Who's right? I hear zero and one is that. 10? Are we doing more binary jokes? Seven, total.

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Speaker 9 15:00

The that tag is different. You did

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Speaker 10 15:08

open SSL. There's one. Product

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Speaker 1 15:32

I'm trying to help people.

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Speaker 4 15:43

Why does core runtime showing open SSL? Because there's a version with and without open SSL. Great question. Core runtime is a weird image. Sometimes you feel like in that sometimes, yeah, yeah, there's no non, there's no Yeah, right. You need, you need cryptography, enforce, right? Yeah, good question. Okay, you all want to do more. Are we having fun? I'm having fun. I've never done this before, so thank you for your patience. Okay, all right, one more open JDK, find 21 run time, number of CVS and streams, not very streams. You're He's fast. Yes, I know. I know you're right. Yeah, this guy's good. This guy gets a free sticker. Okay, now, do you want to hear the coolest thing about the website, all the data that populates this is an API that is open and everybody can access. It's all machine readable. It just populates the UI. This is for humans. It pains me to say this, but after writing, after a decade of writing 1000s of container files, I no longer write container files. Can I admit that? Is this a safe place? I say for unit files, I don't write those either from scratch, I do generate them now with Claude, cursor, whatever, and then I go proofread and tailor and tweak and customize. So if you go, if you'll notice, on the top is this, in the lab the skills,

 Speaker 1 17:34
if it's on the page, it's on the skills, all right,

 Speaker 4 17:37
there's a link on the top for skills. You can point whatever at, and it will generate using our APIs and all the best practices that are in the docs, which is

 Speaker 11 17:56
amazing Anyway,

 Speaker 1 17:59
where is Jeff done? So one thing Ben didn't say, which is actually my favorite

 Speaker 4 18:04
thing.

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Speaker 1 18:05

If you go to the CBE feed, you'll see two numbers on the left hand side of that page. That's actually my favorite page we got that shows the last time we built an image and pushed it to the registry, and every time I opened that page, that number is different, I've caught it hasn't been under three hours since Monday. So this is proof of life, right? This is we are actually honest to goodness building. As soon as we see changes in projects as soon as we can get that code through the pipeline, it's getting the page. Our average build time still stays, staying under an hour exactly where we want it to be. Okay? So the rest of today will be done in that web interface. If you haven't logged in, I'll show you that again. If you have we'll move on. So the way this lays out right we've got tabs across the top here that will say things like kernel console instructions pane on the left hand side, things in the little highlighted boxes are usually commands to run, copy them out, type them by hand, or they might just be files we would like you to read, but it might be a little bit easier to read a snippet there that would be on the journal. Here you can look around do what you want to do. Modules are listed up the top, if you put that little amber looking thing next to the house. That's where you can select module one versus module two. If you want to jump ahead to module two and then go back to module one, just do whatever it tells you to do. Tosin myself and Josh will be around. If you have questions, raise a hand and let

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Speaker 4 19:57

me know. If you want stickers. These are all that's left on planet Earth. Okay, thank you. Sure two of

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Speaker 1 20:05

each. Well, as

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Speaker 8 20:06

long as there's plenty for everyone, yes, we'll do one of each real quick. Make sure

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Speaker 12 20:11

we have enough.



Speaker 6 20:28

Sorry, front bias in a pinch, I



Speaker 8 20:50

day replication, container development.